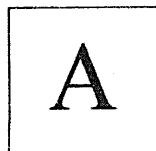
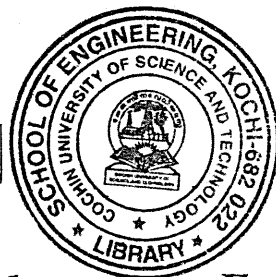


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B.Tech. Degree VII Semester Special Supplementary Examination March 2023

ME 19-205-0701 REFRIGERATION AND AIR CONDITIONING (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Understand the basics of different refrigeration cycles.
 CO2: Study the vapour compression and vapour absorption refrigeration cycles- both theoretical and practical cycles, properties of refrigerants, and selection criteria for refrigerants.
 CO3: Understand the different components of refrigeration systems.
 CO4: Get knowledge of psychrometry, air-conditioning systems and will be able to do the cooling load calculation

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer **ALL** questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a) Explain the term "Tonne of refrigeration".		3	L2	1	1
	(b) What you mean by natural refrigeration and artificial refrigeration? Give two examples each.		3	L2	1	1
	(c) List the difference between vapour compression and vapour absorption refrigeration system.		3	L2	2	1
	(d) Explain the difference between single and multi pressure vapour compression refrigeration system.		3	L1	2	1
	(e) Define the basic principle behind the working of capillary tube as an expansion device.		3	L2	3	1
	(f) Discuss the relative advantages and disadvantages of air cooled and water-cooled condensers.		3	L2	3	1
	(g) What is specific humidity and relative humidity?		3	L2	4	1
	(h) Define the term 'bypass' factor used for cooling or heating coil and find an expression for the same.		3	L1	4	1

PART B

(4 × 12 = 48)

II.	Capacity of refrigerator is 45 tonnes when working between -15°C and 30°C. Find out the mass of ice produced within 24 hours when water is supplied when water is supplied at 20°C. Also find out the minimum power required. Assume machine should be working on Carnot cycle. Latent heat of ice is 335 KJ/Kg.	12	L4	1	2
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OR

III.	(a) Draw the schematic of Bell- Coleman cycle of air refrigeration system, and show the cycle on T-S diagram and P-V diagram.	6	L1	1	1
	(b) Describe briefly the working of pulse tube refrigeration with neat sketch.	6	L1	1	1

(P.T.O.)

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		Marks	BL	CO	PO																			
IV.	(a) Explain the vapour absorption refrigeration system with neat sketch.	6	L1	2	1																			
	(b) Describe the chemical properties of refrigerants.	6	L1	2	1																			
OR																								
V.	A vapour compressor refrigerator uses methyl chloride (R40) and operates between temperature limit of -10°C and 45°C . At the entry to the compressor the refrigerant is dry saturated and after the compressor it acquires the temperature of 60°C . Find the COP of the refrigerator.	12	L4	2	2																			
<table border="1" style="margin: auto; border-collapse: collapse; text-align: center;"> <tr> <th rowspan="2">Saturation temperature in $^{\circ}\text{C}$</th> <th colspan="2">Enthalpy in kJ/kg</th> <th colspan="2">Entropy in kJ/kg K</th> </tr> <tr> <th>Liquid</th> <th>Vapour</th> <th>Liquid</th> <th>Vapour</th> </tr> <tr> <td>-10</td> <td>45.4</td> <td>460.7</td> <td>0.183</td> <td>1.637</td> </tr> <tr> <td>45</td> <td>133.0</td> <td>483.6</td> <td>0.485</td> <td>1.587</td> </tr> </table>						Saturation temperature in $^{\circ}\text{C}$	Enthalpy in kJ/kg		Entropy in kJ/kg K		Liquid	Vapour	Liquid	Vapour	-10	45.4	460.7	0.183	1.637	45	133.0	483.6	0.485	1.587
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45	133.0	483.6	0.485	1.587																				
VI.	(a) With the support of diagrams, briefly explain the different psychrometric process.	6	L1	3	1																			
	(b) Discuss different types of compressor and condensers used for refrigeration.	6	L1	3	1																			
OR																								
VII.	(a) Explains any two rotary compressors.	6	L1	3	1																			
	(b) Define volumetric efficiency of a compressor. What are the factors influencing volumetric efficiency?	6	L1	3	1																			
VIII.	200 m^3 of air per min is passing through the adiabatic humidifier. The condition of air at inlet is 40°C dry bulb temperature and 15% relative humidity and the outlet condition is 25°C dry bulb temperature and 20°C wet bulb temperature. Find the dew point temperature and the amount of water vapour added to the air per minute.	12	L4	4	2																			
OR																								
IX.	The atmospheric air at 30°C DBT 75% RH enters a cooling coil at a rate of $200\text{ m}^3/\text{min}$. The coil dew point temperature is 14°C and by-pass factor of coil is 0.1. Determine:	12	L4	4	2																			
	(i) The temperature of the air leaving the cooling coil.																							
	(ii) The capacity of the cooling coil in tonnes of refrigeration and in kilowatt.																							
	(iii) Amount of water vapour removed per minute.																							
	(iv) Sensible heat factor for the process.																							

Bloom's Taxonomy Levels

L1 = 50%, L2 = 30%, L4 = 20%
